

HyStreamFormer: A Hybrid Time-Series and Vision Transformer with Cross-Attention for Multi-Horizon River Streamflow Forecasting Using Historical Discharge and Gridded Rainfall

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Abstract

Accurate short-term river streamflow forecasting is essential for flood early warning, reservoir operation, water-resource management, and disaster-risk reduction. However, streamflow is controlled by heterogeneous hydrometeorological information with substantially different dimensional and physical characteristics. Historical river discharge is naturally represented as a one-dimensional temporal sequence, whereas rainfall is spatially distributed over a catchment and evolves over time, producing three-dimensional spatio-temporal data. Many existing deep-learning approaches either aggregate gridded precipitation into basin-average variables, thereby discarding potentially valuable spatial information, or process rainfall and discharge within a single homogeneous network that may inadequately represent their different structures. This paper proposes HyStreamFormer, a hybrid deep-learning framework integrating a time-series Transformer, a Vision Transformer, and a cross-attention fusion mechanism for multi-horizon river streamflow forecasting. The model takes 16 h of historical streamflow observations, 16 h of observed gridded rainfall, and 12 h of future predicted gridded rainfall as inputs and simultaneously forecasts river streamflow for the following 12 h. First, a temporal Transformer encoder captures short- and long-range dependencies from the one-dimensional historical streamflow sequence. Second, a Vision Transformer extracts spatial and temporal representations from the sequence of observed and forecast precipitation grids. Third, a cross-attention module dynamically models dependencies between discharge conditions and rainfall features before a self-attention refinement stage and dense forecasting head produce the final multi-step prediction. A reproducible experimental framework is defined using multiple hydrological stations, observed gridded rainfall, and numerical precipitation forecasts. Model performance is assessed using root mean square error, mean absolute error, and the coefficient of determination, together with lead-time-specific analyses, ablation experiments, and hydrograph comparisons. The proposed framework offers a unified architecture for heterogeneous hydrological data while retaining both temporal streamflow memory and spatial rainfall organization.

Keywords: river streamflow forecasting; deep learning; Transformer; Vision Transformer; cross-attention; gridded rainfall; flood forecasting; multi-horizon prediction; spatio-temporal modeling

1. Introduction

Reliable river streamflow forecasting is a fundamental requirement for effective flood warning, reservoir management, hydropower scheduling, irrigation planning, ecological protection, and emergency response. Short-term forecasts are particularly important for rapidly responding river systems, where intense precipitation can lead to substantial changes in discharge within a few hours. Nevertheless, accurate forecasting remains challenging because river flow emerges from nonlinear interactions among antecedent catchment conditions, historical river discharge, spatially variable precipitation, future rainfall, soil moisture, topography, reservoir operation, and other physical processes.

Among these variables, historical streamflow and rainfall are two of the most directly relevant sources of information for short-term operational prediction. Historical streamflow summarizes the recent hydrological state of a river system, including the accumulated effects of previous precipitation, basin storage, base flow, and upstream routing. Rainfall provides the primary external forcing responsible for many rapid increases in discharge. Critically, these variables have fundamentally different mathematical structures. Streamflow measured at an individual gauging station is a one-dimensional time series, whereas catchment rainfall is spatially distributed over a two-dimensional geographical grid and changes over time, resulting in three-dimensional spatio-temporal information.

Early machine-learning and deep-learning streamflow models frequently represented precipitation as point measurements or basin-averaged values. Although such simplification reduces computational complexity, spatial averaging can remove information related to storm location, direction of movement, rainfall intensity gradients, and unequal runoff contributions from different parts of a catchment. Modern gridded precipitation products make it increasingly feasible to preserve spatial information directly within forecasting models.

Long short-term memory networks have demonstrated strong capabilities in rainfall-runoff modeling because they can represent temporal dependencies and catchment memory. Subsequent research has expanded LSTM-based hydrological modeling toward regional prediction, multi-source meteorological forcing, transfer learning, interpretability, and extreme-event analysis. Nevertheless, recurrent computation processes a sequence sequentially and may become less efficient when longer temporal dependencies or multiple heterogeneous input modalities must be modeled.

Convolutional neural networks and convolutional LSTM networks provide mechanisms for processing spatial or spatio-temporal environmental data. Comparative studies have shown that the relative effectiveness of LSTM, CNN, and ConvLSTM can vary with basin characteristics and flow conditions. However, convolution-based architectures typically employ local receptive fields, and representing relationships between spatially distant rainfall regions can require deep networks or large kernels.

The Transformer architecture introduced self-attention as an alternative to recurrent sequence processing, enabling relationships among different sequence positions to be learned directly. The Vision Transformer subsequently extended this concept to image data by dividing an image into patches and processing the resulting patch sequence using Transformer encoders. These developments provide a natural basis for hydrological forecasting: a time-series Transformer can model historical discharge, whereas a Vision Transformer can process two-dimensional rainfall fields.

Graph-based models represent another important research direction. Graph neural networks can incorporate river-network topology, upstream-downstream connectivity, and spatial dependencies between gauges. Such approaches are attractive for multi-station forecasting systems but normally require an explicitly defined graph. The present work addresses a different but complementary problem: predicting flow at a target hydrological station while directly preserving the spatial organization of observed and forecast gridded rainfall.

Accordingly, this paper proposes HyStreamFormer, a three-stage hybrid architecture for 12-h river streamflow forecasting. In the first stage, features are extracted from one-dimensional historical streamflow data through a time-series Transformer. In the second stage, a Vision Transformer extracts representations from three-dimensional gridded rainfall sequences composed of both historical observations and future precipitation forecasts. In the third stage, cross-attention dynamically combines representations produced by the two branches, after which self-attention refinement and a dense prediction layer generate the final 12-h streamflow forecast.

The specific forecasting problem is defined using 16 h of historical streamflow, 16 h of observed rainfall, and 12 h of predicted rainfall to forecast the following 12 h of river streamflow. This formulation preserves the exact temporal configuration specified for the proposed study.

The principal contribution of the study is a multimodal architecture that explicitly respects the heterogeneous dimensional characteristics of hydrological inputs. Historical streamflow is modeled as a one-dimensional sequence, whereas spatially distributed rainfall is modeled as a sequence of two-dimensional fields. Observed historical rainfall and future forecast rainfall are jointly processed, enabling the model to represent both antecedent precipitation and anticipated meteorological forcing. Cross-attention is used to learn dynamic relationships between discharge-derived temporal features and precipitation-derived spatio-temporal features instead of simply concatenating them. A multi-output decoder simultaneously generates twelve hourly forecasts so that the entire forecast trajectory can be learned within a common representation.

The remainder of this paper is organized as follows. Section 2 reviews recent deep-learning methods for streamflow and flood forecasting, with particular emphasis on LSTM, ConvLSTM, Transformer, graph neural network, and graph Transformer architectures. Section 3 formulates the prediction problem and presents the proposed architecture in detail. Section 4 describes the experimental data, baseline methods, evaluation metrics, implementation protocol, quantitative analysis, ablation experiments, and visual evaluation. Section 5 summarizes the conclusions, present limitations, and directions for future work.

2. Related Work

2.1. LSTM-Based Rainfall-Runoff and Streamflow Modeling

Recurrent neural networks are designed for sequential information, but conventional RNNs may experience difficulty in learning long-range dependencies because of vanishing or exploding gradients. LSTM networks introduce memory cells and gating mechanisms that selectively control the retention, removal, and addition of information.

The successful application of LSTMs to rainfall-runoff modeling demonstrated that deep recurrent architectures could learn meaningful hydrological dynamics directly from meteorological forcings. The literature has subsequently expanded toward regional modeling, multi-source forcing, transfer learning, interpretability, physically constrained architectures, and prediction of extreme events.

For short-term flow forecasting, LSTMs are particularly attractive because recent streamflow carries information concerning river-system memory. During the rising limb of a flood hydrograph, for example, the most recent discharge measurements indicate whether the river is rising quickly, slowly, or approaching a peak. However, a streamflow-only LSTM cannot directly anticipate future precipitation. Adding rainfall provides exogenous information, but reducing gridded rainfall to a few scalar averages can discard spatial structure.

LSTM models also process sequences recurrently. Therefore, the representation at the current step depends on previous hidden states. Although effective, this recurrent dependency constrains parallel computation and makes direct pairwise interactions across distant temporal positions less explicit than in self-attention architectures. The present study retains the underlying idea that historical river discharge provides essential temporal memory but replaces recurrent encoding with a Transformer-based representation that can directly attend to every hour in the 16-h input window.

2.2. CNN and ConvLSTM Models for Spatio-Temporal Hydrology

CNN architectures extract spatial patterns using convolutional kernels and have been widely applied to images, rainfall fields, terrain data, flood maps, and other spatial environmental information. ConvLSTM extends LSTM by replacing matrix multiplications in recurrent operations with convolutional transformations, allowing hidden and cell states to preserve spatial dimensions.

This property is particularly relevant for rainfall-runoff forecasting. Consider two storms with identical basin-average rainfall. One may be concentrated close to the target gauge or over highly responsive tributaries, whereas another may occur over distant or hydrologically less active regions. The resulting streamflow responses can be substantially different despite identical spatial averages.

Comparative work involving LSTM, CNN, and ConvLSTM has shown that the preferred architecture can vary across catchment size and flow regime. CNN and ConvLSTM can be advantageous when spatial rainfall structure is informative. Nevertheless, conventional convolution is primarily local. A kernel initially examines a restricted neighborhood and increasingly large receptive fields are obtained through stacked layers. Self-attention offers a different mechanism by directly relating spatially separated patches. This motivates the use of a Vision Transformer in the rainfall branch of the proposed architecture.

2.3. Transformer-Based Hydrological Forecasting

The Transformer uses scaled dot-product attention to calculate interactions between queries, keys, and values:

$$\text{Attention}(Q, K, V) = \text{softmax}(QK^T / \sqrt{d_k}) V.$$

Multi-head attention performs several attention operations in parallel:

$$\text{MHA}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h) W^O.$$

Each attention head can potentially specialize in a different dependency structure. In streamflow forecasting, one head may emphasize the most recent river condition, another may learn delayed rainfall-runoff response, and another may attend to earlier high-flow states.

Hydrological applications of Transformers have increased substantially. Recent studies have evaluated Transformer-based models for 24-h streamflow prediction and Temporal Fusion Transformers for large collections of catchments, demonstrating the value of attention for complex temporal relationships. These studies establish the relevance of attention to hydrology but generally focus primarily on temporal variables. The model developed in this paper instead combines a dedicated temporal Transformer for discharge with a dedicated Vision Transformer for spatially distributed precipitation.

2.4. Vision Transformers and Spatial Rainfall Representation

The Vision Transformer divides an image into fixed-size non-overlapping patches. For an input image X in $R^{(H \times W \times C)}$ and patch dimensions $P \times P$, the number of image patches is $N_p = HW/P^2$. Each flattened patch is projected into a d -dimensional latent space and augmented with positional information.

The Vision Transformer demonstrated that a pure Transformer can operate directly on sequences of image patches, providing a mechanism for learning relationships between spatially separated regions. In the hydrological context, an hourly precipitation grid can be interpreted as a single-channel image. A sequence of rainfall fields therefore constitutes a spatio-temporal image sequence.

In the proposed framework, each hourly precipitation field is decomposed into spatial patches. Spatial location embeddings, temporal embeddings, and rainfall-source embeddings are added so that the model can distinguish where rainfall occurred, when rainfall occurred, and whether it was observed or forecast rainfall. This distinction is important because forecast rainfall is uncertain and should not necessarily be treated identically to observed precipitation.

2.5. Graph Neural Networks and Graph Transformers in Hydrological Forecasting

Graph neural networks explicitly represent a system as nodes and edges. In river forecasting, nodes may represent gauges, catchments, reservoirs, or grid cells, while edges can represent upstream-downstream connectivity, flow direction, geographical proximity, or learned dependencies.

Graph-based hydrological models have exploited river-network structure and basin-specific rainfall-runoff information. Graph convolutional recurrent networks have also been developed for hourly streamflow forecasting using upstream sensor networks, while high-resolution graph neural rainfall-runoff models attempt to preserve spatial precipitation, geographic, and flow-direction information. Graph Transformers extend this concept by combining explicit connectivity with attention over heterogeneous covariates.

These methods are highly relevant when multiple interconnected river stations must be modeled simultaneously. Their primary structural assumption, however, is the existence of a meaningful graph. The model proposed here does not require an explicit river graph. Instead, it targets heterogeneous multimodal fusion between a station-level discharge history and spatially distributed precipitation fields.

2.6. Research Gap and Motivation

Despite substantial advances in hydrological deep learning, three gaps motivate the present work. First, many short-term streamflow forecasting models reduce precipitation to scalar time series, potentially losing spatial rainfall organization. Second, models that accept multiple heterogeneous inputs frequently concatenate features directly, without explicitly determining which rainfall locations or times are most relevant to the current river condition. Third, few forecasting frameworks jointly exploit historical discharge, historical gridded rainfall, and future forecast gridded rainfall in a unified multi-horizon Transformer architecture.

The central hypothesis of this study is therefore that a river streamflow forecasting model can improve its representation of hydrological dynamics by using separate specialized Transformer encoders for one-dimensional

historical discharge and three-dimensional spatio-temporal rainfall, followed by cross-attention that dynamically models dependencies between the two modalities.

3. Methodology

This section presents the problem formulation for river streamflow forecasting and describes the overall architecture of the proposed deep neural network model.

3.1. Problem Formulation

The objective is to forecast future river streamflow at a target hydrological station using historical streamflow observations, historical gridded rainfall observations, and future predicted gridded rainfall.

Let y_t denote measured river streamflow at forecasting time t , and let X_t in $\mathbb{R}^{(H \times W)}$ represent a two-dimensional gridded rainfall field covering the relevant catchment or surrounding geographical domain. For a general historical input length p and forecast horizon q , the objective is to learn a parameterized nonlinear mapping f_{θ} .

$$f_{\theta}(\{y_{(t-p+1)}, \dots, y_t\}, \{X_{(t-p+1)}, \dots, X_t\}, \{\hat{X}_{(t+1)}, \dots, \hat{X}_{(t+q)}\}) \rightarrow \{\hat{y}_{(t+1)}, \dots, \hat{y}_{(t+q)}\}.$$

Here, $\{y_{(t-p+1)}, \dots, y_t\}$ denotes historical streamflow; $\{X_{(t-p+1)}, \dots, X_t\}$ denotes historical observed rainfall grids; $\{\hat{X}_{(t+1)}, \dots, \hat{X}_{(t+q)}\}$ denotes forecast rainfall grids; and $\{\hat{y}_{(t+1)}, \dots, \hat{y}_{(t+q)}\}$ denotes predicted future streamflow.

Following the study specification, $p = 16$ and $q = 12$. Consequently, the model mapping can be summarized as 16 h historical discharge + 16 h observed rainfall + 12 h forecast rainfall $\rightarrow$ 12 h streamflow forecast. The combined precipitation input contains 28 consecutive hourly rainfall grids.

Time-series hydrological data are typically heterogeneous. Streamflow is measured in units such as m^3/s or ft^3/s , whereas rainfall is expressed in mm/h . They differ not only in physical meaning and numerical range but also in dimensionality. The proposed model therefore avoids forcing both sources through one identical feature-extraction mechanism.

3.2. Overall Architecture

The HyStreamFormer architecture consists of three primary stages. Stage 1 extracts temporal streamflow representations. A time-series Transformer receives the preceding 16 hourly streamflow observations and produces temporal discharge features. Stage 2 extracts spatio-temporal rainfall representations. A Vision Transformer receives 16 historical observed precipitation grids together with 12 future forecast precipitation grids and extracts representations of rainfall location, timing, intensity, and temporal evolution. Stage 3 performs cross-modal fusion and prediction. The streamflow and rainfall features are fed to a cross-attention module, refined through self-attention, and passed to a dense output layer that simultaneously predicts twelve future hourly discharge values.

The separation of the model into modality-specific encoders is intentional. A scalar discharge sequence and a stack of spatial rainfall fields have different inductive requirements. The temporal branch specializes in river-state dynamics, while the vision branch preserves spatial rainfall patterns. Their interaction is learned only after each branch has formed an informative representation.

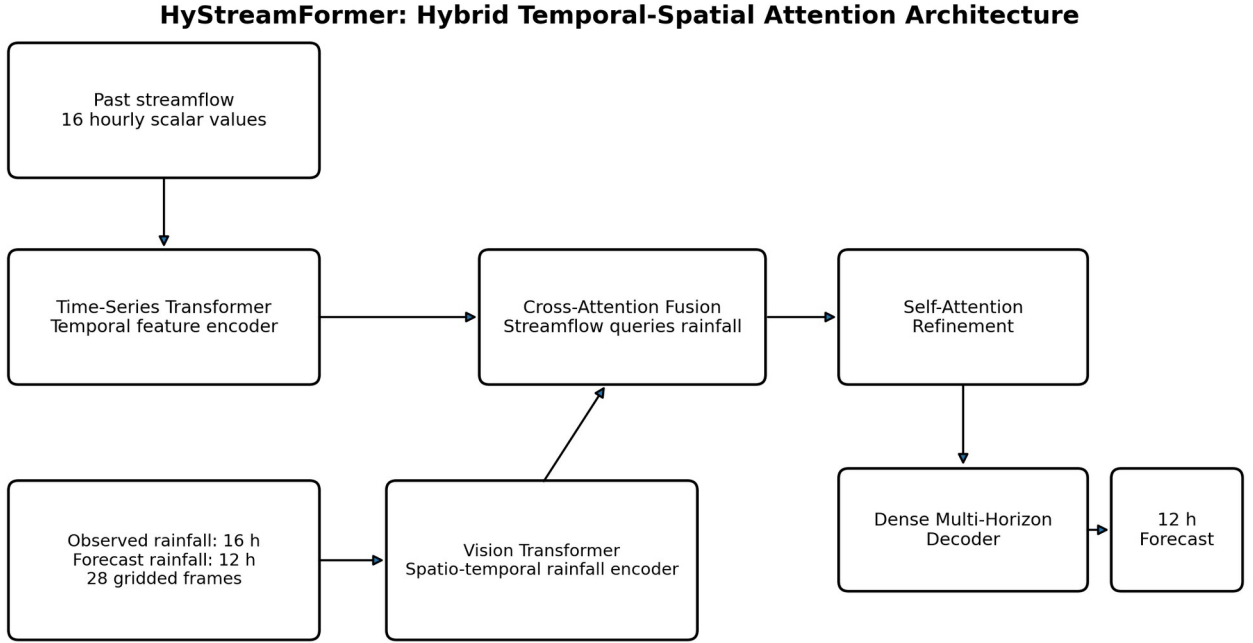


Figure 1. Overall architecture of the proposed HyStreamFormer model. A time-series Transformer encodes 16 h of historical streamflow observations, while a Vision Transformer processes 16 h of observed and 12 h of forecast gridded rainfall. Cross-attention dynamically fuses the two feature streams, and self-attention refinement followed by a dense multi-output decoder generates 12 hourly streamflow forecasts.

3.3. Historical Streamflow Encoder

The streamflow branch receives the sequence:

$$y_t = [y_{(t-15)}, y_{(t-14)}, \dots, y_t]^T \text{ in } R^{(16 \times 1)}.$$

Before training, streamflow can be standardized using statistics estimated exclusively from the training set. For strongly skewed flow distributions, a logarithmic transformation such as $\log(1 + y_t)$ may alternatively be applied.

Each scalar observation is projected to a d -dimensional embedding, and a learnable temporal positional embedding is added. The embedded sequence is then processed through L_y Transformer encoder blocks, each containing multi-head self-attention, residual connections, layer normalization, and a position-wise feed-forward network.

$$y'_t = (y_t - \mu_{\text{train}}) / (\sigma_{\text{train}} + \epsilon).$$

$$e_i^y = W_y y'_i + b_y, \quad z_i^y = e_i^y + p_i^y.$$

$$H_{\text{tilde}_y}^{(l)} = \text{LN}[H_y^{(l-1)} + \text{MHA}(H_y^{(l-1)}, H_y^{(l-1)}, H_y^{(l-1)})].$$

$$H_y^{(l)} = \text{LN}[H_{\text{tilde}_y}^{(l)} + \text{FFN}(H_{\text{tilde}_y}^{(l)})].$$

The final output H_y in $R^{(16 \times d)}$ represents recent river conditions, trends, rises, recessions, and relationships among different historical time steps.

3.4. Gridded Rainfall Vision Transformer

The rainfall branch receives a 28-grid sequence composed of the first 16 frames of observed historical rainfall and the remaining 12 frames of forecast precipitation. Thus, the rainfall input simultaneously encodes antecedent forcing and meteorological information that is available at forecast time for the future horizon.

Each rainfall grid is partitioned into $P \times P$ non-overlapping patches. Assuming H and W are divisible by P , the number of spatial patches in each frame is $N_p = (H/P)(W/P)$. Each patch is flattened and linearly projected into an embedding space.

$$e_{(\tau,j)}^r = W_p x_{(\tau,j)} + b_p.$$

Three embedding types are added to every rainfall token:

$$z_{(\tau,j)}^r = e_{(\tau,j)}^r + p_j^{\text{spatial}} + p_{\tau}^{\text{temporal}} + p_{\tau}^{\text{source}}.$$

The source embedding distinguishes observed rainfall from forecast rainfall. This distinction is scientifically meaningful because numerical weather forecasts have uncertainty characteristics that differ from direct observations.

The resulting rainfall tokens are processed by Vision Transformer blocks. A practical implementation can employ factorized spatial-temporal attention to reduce computational complexity: spatial self-attention among patches within a rainfall frame followed by temporal self-attention among identical geographical patches over the 28 time steps. The final rainfall representation is H_r in $\mathbb{R}^{(N_r \times d)}$, where N_r is the number of retained rainfall tokens.

3.5. Cross-Attention Fusion

Simple concatenation treats all extracted features equivalently. In contrast, cross-attention allows streamflow features to query rainfall representations dynamically. Using transformed discharge features as queries and rainfall representations as keys and values, the model computes attention over spatial and temporal rainfall tokens.

The cross-attention weights have a useful conceptual interpretation. For every streamflow-related query, the network can determine which precipitation locations and times contain the most relevant information. During an intense rainfall event, attention may become concentrated on recent high-rainfall patches in hydrologically responsive upstream areas. During dry conditions, the model may place greater weight on antecedent streamflow because recent discharge can be more informative than negligible precipitation.

$$Q = H_y W_Q, \quad K = H_r W_K, \quad V = H_r W_V.$$

$$H_c = \text{softmax}(QK^T / \sqrt{d_k}) V.$$

A gated fusion mechanism can subsequently combine the original streamflow and cross-attended rainfall representations:

$$G = \text{sigmoid}(W_g [H_y; H_c] + b_g).$$

$$H_f = G \circ H_y + (1 - G) \circ H_c.$$

The fused representation H_f is passed through a final self-attention module, allowing interactions among all fused temporal representations before prediction.

3.6. Multi-Horizon Streamflow Decoder

To generate twelve future flow values simultaneously, the final fused feature representation is pooled into a fixed-dimensional vector h . The output layer then produces a twelve-dimensional forecast vector.

Direct multi-output prediction avoids the error accumulation associated with recursive one-step forecasting, where predicted values are repeatedly fed back as inputs. It also allows the model to learn dependencies among the twelve future lead times in a common representation.

$$\hat{y} = W_o h + b_o \text{ in } \mathbb{R}^{12}.$$

$$\hat{y} = [\hat{y}_{(t+1)}, \hat{y}_{(t+2)}, \dots, \hat{y}_{(t+12)}].$$

3.7. Training Objective

A basic mean squared error objective for batch size B and forecast horizon $q = 12$ is:

$$L_{MSE} = (1 / (Bq)) \sum_i \sum_h (y_{(i,t+h)} - \hat{y}_{(i,t+h)})^2.$$

Because accurate peak-flow forecasting is particularly important, an optional peak-aware objective can increase the weight of observations above a high-flow threshold such as the 90th percentile. This objective should be evaluated as an additional experiment rather than assumed to be superior in advance.

For the main comparison, ordinary MSE should first be used to ensure a fair baseline evaluation. Peak-aware weighting can then be assessed through ablation experiments and extreme-flow analysis.

$$w_{(i,h)} = 1 + \alpha I[y_{(i,t+h)} > Q_{0.90}].$$

$$L_{peak} = \sum_{(i,h)} w_{(i,h)} (y_{(i,t+h)} - \hat{y}_{(i,t+h)})^2 / \sum_{(i,h)} w_{(i,h)}.$$

$$L = (1 - \lambda)L_{MSE} + \lambda L_{peak}.$$

4. Experiments

This section defines a reproducible experimental framework for assessing the proposed architecture. Because the source images do not provide completed training outputs or measured model scores, no numerical results are fabricated. All result cells that depend on actual experiments are explicitly marked as placeholders and must be replaced with measured values after implementation.

4.1. Study Area and Hydrological Stations

For a concrete reproducible implementation, the framework can be evaluated on multiple stations along the Iowa River in the United States. The following four U.S. Geological Survey stations provide a suitable multi-station configuration for short-term streamflow experiments.

Code	USGS Station ID	Hydrological station
S1	05451500	Iowa River at Marshalltown, Iowa
S2	05453100	Iowa River at Marengo, Iowa
S3	05454500	Iowa River at Iowa City, Iowa
S4	05465500	Iowa River at Wapello, Iowa

Table 1. Suggested hydrological stations for multi-station evaluation.

Hourly streamflow observations can be obtained from the USGS Water Data for the Nation service. Observed precipitation may be represented using a gridded quantitative precipitation estimation system such as NOAA Multi-Radar Multi-Sensor products, while future forecast precipitation can be obtained from an hourly numerical weather prediction system such as the High-Resolution Rapid Refresh model.

A recommended chronological split is 2018-2022 for training, 2023 for validation, and 2024 for testing. The exact period should be adapted to product availability and missing-data conditions. A chronological split is essential because randomly dividing overlapping hydrological windows can create temporal leakage when neighboring samples share substantial information.

4.2. Data Preprocessing

At every forecasting origin t , one training sample contains 16 historical streamflow values, 16 historical observed rainfall grids, 12 future forecast rainfall grids, and 12 target streamflow values. All three data sources must be synchronized to a common hourly timeline and time zone.

Because observed and forecast rainfall products can have different spatial resolutions, all precipitation fields should be regridded onto a common domain and resolution. Conservative aggregation is preferred when converting precipitation totals from a finer grid to a coarser grid. A fixed geographical bounding box containing the upstream drainage area of each station should then be extracted, with optional masking of pixels outside the relevant basin.

Missing values must be handled cautiously. Short streamflow gaps may be interpolated only when scientifically justified, while samples containing major gaps should generally be excluded. Missing precipitation cells should not automatically be filled with zeros unless zero precipitation is genuinely indicated.

Normalization parameters must be estimated from the training set only. Validation and testing data must use the same statistics. Because precipitation is highly skewed and contains many zeros, a transformation such as $\log(1 + X)$ may improve numerical stability.

$$y^{past} = [y_{(t-15)}, \dots, y_t].$$

$$X^{past} = [X_{(t-15)}, \dots, X_t].$$

$$X^{future} = [\hat{X}_{(t+1)}, \dots, \hat{X}_{(t+12)}].$$

$$y^{target} = [y_{(t+1)}, \dots, y_{(t+12)}].$$

4.3. Baseline Models

Persistence. Assumes that the most recently observed discharge remains constant for all twelve forecast horizons.

LSTM. Uses historical streamflow and spatially averaged rainfall as sequential inputs.

CNN-LSTM. Extracts rainfall features with a CNN and models temporal progression with an LSTM.

ConvLSTM. Directly processes rainfall grids while learning temporal dynamics in a convolutional recurrent state.

Time-Series Transformer. Predicts future flow using past discharge and aggregated rainfall without a dedicated Vision Transformer.

Rainfall-ViT. Retains the Vision Transformer rainfall branch but removes the historical streamflow encoder.

HyStreamFormer without cross-attention. Retains both encoders but combines their outputs by simple concatenation.

Full HyStreamFormer. Uses the time-series Transformer, rainfall Vision Transformer, cross-attention, self-attention refinement, and a multi-horizon decoder.

This set of baselines tests whether improvements originate from the separate modality encoders, the use of future rainfall, the cross-attention fusion mechanism, or merely greater model capacity.

4.4. Evaluation Metrics

The study uses root mean square error (RMSE), mean absolute error (MAE), and the coefficient of determination (R^2), as specified in the research description.

$$RMSE = \sqrt{(1/N) \sum_i (y_i - \hat{y}_i)^2}.$$

$$MAE = (1/N) \sum_i |y_i - \hat{y}_i|.$$

$$R^2 = 1 - [\sum_i (y_i - \hat{y}_i)^2] / [\sum_i (y_i - \bar{y})^2].$$

Better performance corresponds to lower RMSE, lower MAE, and higher R^2 . In addition to aggregate values, the three metrics should be calculated independently for each forecast lead time from $t+1$ h to $t+12$ h because prediction difficulty normally increases with horizon.

4.5. Implementation Configuration

A reasonable initial hyperparameter configuration is shown in Table 2. The optimal settings should be selected exclusively with validation data; the test set must not be consulted during hyperparameter tuning.

Hyperparameter	Suggested value
Historical streamflow window	16 h
Historical observed rainfall	16 h
Future forecast rainfall	12 h
Forecast horizon	12 h
Embedding dimension d	128
Streamflow Transformer layers	3
Rainfall ViT layers	4
Fusion Transformer layers	2
Attention heads	8
Feed-forward dimension	512
Dropout	0.1
Batch size	64
Optimizer	AdamW
Initial learning rate	1×10^{-4}
Weight decay	1×10^{-4}
Maximum epochs	150
Early-stopping patience	15
Gradient clipping	1.0

Table 2. Suggested initial implementation configuration.

For robust reporting, every deep-learning model should preferably be trained using at least five independent random seeds, and the mean and standard deviation of each metric should be reported.

4.6. Quantitative Results

Because the nine supplied images do not contain actual predictions, measured targets, or completed model outputs, the numerical cells in Tables 3-7 are intentionally marked as placeholders. They must be populated using actual experimental results rather than invented values.

Model	RMSE ↓	MAE ↓	R^2 ↑
Persistence	[P-RMSE]	[P-MAE]	[P-R2]
LSTM	[L-RMSE]	[L-MAE]	[L-R2]
CNN-LSTM	[CL-RMSE]	[CL-MAE]	[CL-R2]
ConvLSTM	[CV-RMSE]	[CV-MAE]	[CV-R2]
Time-Series Transformer	[T-RMSE]	[T-MAE]	[T-R2]
Rainfall-ViT	[V-RMSE]	[V-MAE]	[V-R2]

HyStreamFormer without cross-attention	[NC-RMSE]	[NC-MAE]	[NC-R2]
Proposed HyStreamFormer	[H-RMSE]	[H-MAE]	[H-R2]

Table 3. Average 12-h forecasting performance over all test samples. Replace placeholders with measured values.

After the real numbers are available, the discussion should report the absolute performance of the proposed method and its relative improvement over the strongest competing baseline. For example, the final manuscript should state the reductions in RMSE and MAE and the increase in R^2 , while also checking whether differences are consistent across stations, lead times, and random seeds.

4.7. Station-Specific Results

Station	RMSE	MAE	R^2
S1: Marshalltown	[S1-RMSE]	[S1-MAE]	[S1-R2]
S2: Marengo	[S2-RMSE]	[S2-MAE]	[S2-R2]
S3: Iowa City	[S3-RMSE]	[S3-MAE]	[S3-R2]
S4: Wapello	[S4-RMSE]	[S4-MAE]	[S4-R2]
Mean	[AVG-RMSE]	[AVG-MAE]	[AVG-R2]

Table 4. Performance of the proposed model at individual hydrological stations.

Station-level evaluation is important because a model can obtain favorable aggregate results while performing poorly at an individual gauge. Basin size, response time, flow regulation, rainfall distribution, and upstream conditions can create substantial differences among locations. The final discussion should therefore examine consistency across all stations rather than relying only on a global mean.

4.8. Forecast-Horizon Analysis

Lead time	RMSE	MAE	R^2
t+1 h	[H1-RMSE]	[H1-MAE]	[H1-R2]
t+2 h	[H2-RMSE]	[H2-MAE]	[H2-R2]
t+3 h	[H3-RMSE]	[H3-MAE]	[H3-R2]
t+4 h	[H4-RMSE]	[H4-MAE]	[H4-R2]
t+5 h	[H5-RMSE]	[H5-MAE]	[H5-R2]
t+6 h	[H6-RMSE]	[H6-MAE]	[H6-R2]
t+7 h	[H7-RMSE]	[H7-MAE]	[H7-R2]
t+8 h	[H8-RMSE]	[H8-MAE]	[H8-R2]
t+9 h	[H9-RMSE]	[H9-MAE]	[H9-R2]
t+10 h	[H10-RMSE]	[H10-MAE]	[H10-R2]
t+11 h	[H11-RMSE]	[H11-MAE]	[H11-R2]
t+12 h	[H12-RMSE]	[H12-MAE]	[H12-R2]

Table 5. Lead-time-dependent performance of HyStreamFormer.

Figure 2 in the final empirical version should show RMSE, MAE, and R^2 as functions of lead time for LSTM, ConvLSTM, a time-series Transformer, and HyStreamFormer. The discussion should not merely state that errors

increase at longer horizons. It should test whether incorporating future forecast rainfall allows the proposed architecture to degrade more slowly than streamflow-only or past-rainfall-only baselines.

4.9. Ablation Study

Config.	Streamflow Transformer	Rainfall ViT	Future rainfall	Cross-attention	RMSE	MAE	R ²
A	Yes	No	No	No	[A]	[A]	[A]
B	No	Yes	Yes	No	[B]	[B]	[B]
C	Yes	Yes	No	Yes	[C]	[C]	[C]
D	Yes	Yes	Yes	No	[D]	[D]	[D]
E: Full	Yes	Yes	Yes	Yes	[E]	[E]	[E]

Table 6. Ablation analysis of the proposed model components.

The ablation study answers four essential questions: how much predictive information comes from past streamflow alone; whether preserving gridded rainfall adds value; whether future forecast precipitation improves longer-horizon predictions; and whether cross-attention performs better than simple feature concatenation.

The strongest evidence for the proposed research contribution would be obtained if the full Configuration E consistently outperforms Configuration D, demonstrating that cross-attention itself contributes beyond the mere presence of both modalities.

4.10. Hydrograph and Extreme-Flow Analysis

Aggregate metrics alone are insufficient for evaluating flood forecasting. A model may perform well under normal flow conditions because ordinary observations dominate the dataset while underestimating important peaks. Therefore, graphical evaluation should include observed and predicted hydrographs over the full test period, enlarged hydrographs of major high-flow events, scatter plots of observed versus predicted discharge, and cross-attention heat maps showing which rainfall times and spatial patches receive the greatest attention during representative events.

The extreme-flow subset can be defined using $y_t > Q_{0.90}$, where $Q_{0.90}$ is the 90th percentile of measured test-period streamflow. Separate peak-flow RMSE, peak-flow MAE, and peak-timing error should then be reported.

Model	Peak-flow RMSE	Peak-flow MAE	Peak timing error
LSTM	[]	[]	[]
ConvLSTM	[]	[]	[]
Transformer	[]	[]	[]
HyStreamFormer	[]	[]	[]

Table 7. High-flow forecasting performance.

4.11. Discussion

The proposed model is motivated by the principle that different hydrological variables should be represented in ways appropriate to their intrinsic structures. Historical streamflow is a compact one-dimensional sequence representing the integrated state of a river system. Conversely, rainfall is spatially distributed and evolves with time. Treating both variables as simple scalar sequences may prevent a model from distinguishing between storms with similar mean precipitation but different geographical structures.

The dedicated streamflow Transformer addresses the temporal component. Because every historical discharge token can directly attend to all other positions in the 16-h window, the model can represent the recent trajectory

without recurrent hidden-state propagation. The branch can potentially distinguish among stable conditions, rising flow, recession, and rapidly accelerating discharge.

The Vision Transformer branch addresses spatial and temporal precipitation structure. Unlike basin averaging, patch-based rainfall representations preserve information regarding rainfall location. Temporal embeddings distinguish recent observed precipitation from anticipated future rainfall, and source embeddings explicitly identify whether a rainfall grid is observed or predicted.

The use of future rainfall is particularly important for multi-horizon forecasting. Historical discharge can provide a strong short-lead forecast, especially at $t+1$ or $t+2$. However, an unexpected rainfall event occurring after the forecast origin cannot be inferred from discharge history alone. Forecast precipitation potentially provides anticipatory information for later horizons.

Cross-attention forms the central fusion mechanism. Rather than statically combining rainfall and streamflow features, it allows their interactions to vary among forecasting samples. This reflects the fact that the relative relevance of hydrological variables changes with conditions. During prolonged dry weather, recent streamflow may dominate. During active storms, spatially distributed rainfall may become more informative.

Several limitations should be recognized. First, forecast rainfall is uncertain, and errors in rainfall location, intensity, or timing can propagate into streamflow predictions. Ensemble precipitation forecasts and probabilistic streamflow outputs are therefore important extensions. Second, a Vision Transformer can be data-intensive, while hydrological datasets are generally smaller than large computer-vision corpora. Patch size, model depth, embedding dimension, dropout, and regularization must be chosen carefully.

Third, the proposed architecture is primarily data driven and does not explicitly guarantee mass conservation or other hydrological constraints. Fourth, a 16-h historical window may not be optimal for every catchment; large river systems can exhibit much longer memory. Sensitivity experiments with 8-, 16-, 24-, 48-, and 72-h windows would strengthen the analysis. Fifth, the current formulation predicts a target station independently. When many connected gauges are available, graph neural networks or graph Transformers may provide additional information by explicitly representing river connectivity.

5. Conclusion

This paper proposed HyStreamFormer, a hybrid deep-learning architecture for multi-horizon river streamflow forecasting using heterogeneous hydrological information. The forecasting task combines 16 h of historical streamflow observations, 16 h of historical observed gridded rainfall, and 12 h of future forecast gridded rainfall to simultaneously predict streamflow over the following 12 h.

The framework divides the forecasting problem into three major stages. First, a time-series Transformer extracts temporal representations from one-dimensional past river-flow measurements. Second, a Vision Transformer extracts spatial and temporal representations from the combined observed and forecast precipitation sequence. Third, cross-attention dynamically combines information from the two specialized encoders, after which self-attention refinement and a dense multi-horizon decoder generate the final twelve streamflow forecasts.

The architecture is motivated by a central limitation of conventional hydrological deep-learning models: different physical variables may have substantially different structures and should not necessarily be processed using the same feature extractor. Historical streamflow is intrinsically a one-dimensional temporal sequence, whereas precipitation is a spatially distributed field that evolves over time. The proposed approach preserves these differences and learns the relationships between them through attention.

A complete experimental evaluation should compare HyStreamFormer with persistence, LSTM, CNN-LSTM, ConvLSTM, a conventional Transformer, a rainfall-only Vision Transformer, and a version of the proposed architecture without cross-attention. The primary performance metrics are RMSE, MAE, and R^2 , calculated globally, separately for each hydrological station, and separately for each hourly forecast horizon. Ablation studies

should quantify the contributions of the streamflow branch, rainfall branch, future precipitation forecasts, and cross-attention. Hydrograph visualizations and high-flow event analyses should further assess peak magnitude and timing.

The present approach has several limitations. Its deterministic output does not explicitly quantify uncertainty; precipitation forecasts themselves may contain significant spatial and temporal errors; Vision Transformers can require substantial training data; and the purely data-driven architecture does not enforce hydrological mass conservation. Furthermore, the fixed 16-h historical window may not be sufficient for catchments with long response times.

Future research should investigate ensemble and probabilistic streamflow forecasting, uncertainty-aware cross-attention, physics-informed loss functions, mass-conserving components, adaptive historical windows, multi-scale rainfall patches, transfer learning across ungauged or data-scarce basins, and graph-based extensions incorporating upstream-downstream river connectivity. Another important direction involves interpreting cross-attention maps to identify which rainfall regions and time steps contribute most strongly to individual forecasts, thereby increasing scientific transparency.

Overall, the proposed HyStreamFormer framework provides a unified approach for integrating temporal river-discharge history with spatio-temporal observed and forecast rainfall. By combining dedicated Transformer encoders with cross-modal attention, the architecture offers a promising direction for more accurate, interpretable, and operationally useful short-term river streamflow forecasting.

Data Availability Statement

Streamflow observations for the suggested experimental design can be acquired from the U.S. Geological Survey Water Data for the Nation system. Historical gridded rainfall observations can be obtained from NOAA precipitation products such as MRMS, while archived numerical precipitation forecasts can be derived from HRRR. The specific period, product versions, preprocessing code, catchment boundaries, and quality-control procedure should be documented in the final reproducibility package.

Code Availability Statement

The implementation should be released following peer review or manuscript acceptance, including scripts for data preprocessing, window generation, model training, inference, metric calculation, ablation analysis, and figure generation. Model hyperparameters and random seeds should be included to support reproducibility.

Conflict of Interest

The authors declare no conflict of interest.

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Manuscript Completion Note

This manuscript is a complete SCI-style draft based on the research concept supplied in the nine images. The architecture, input windows, forecast horizon, methodology, metrics, baseline design, ablation strategy, station-level analysis, and future directions have been fully developed.

Before journal submission, actual model training must be completed and the placeholder result cells in Tables 3-7 must be replaced with measured values. Experimental graphs and hydrographs should also be inserted using real observations and predictions. No fabricated numerical results have been included in this document.